

TURKISTAN, Kazakhstan

WMO: 381980

Lat: 43.2719N

Lon: 68.1914E

Elev: 207

StdP: 98.86

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.5	-15.1	-22.0	0.5	-16.9	-18.5	0.7	-13.8	13.2	-2.0	9.6	-2.7	1.0	50	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	39.5	19.4	38.0	18.9	36.6	18.6	20.7	35.2	20.1	34.4	19.5	33.6	3.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.2	23.4	15.3	11.1	23.8	14.1	10.3	24.2	60.8	34.8	58.5	34.3	56.4	33.6	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	7.7	6.2	-20.2	42.3	5.4	1.5	-24.1	43.4	-27.3	44.2	-30.3	45.1	-34.3	46.1		
(5)			-20.4	22.2	5.4	1.2	-24.3	23.1	-27.5	23.8	-30.5	24.5	-34.4	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	-2.5	0.1	8.5	15.4	21.9	26.8	28.4	27.4	21.1	12.8	5.5	-1.3	(6)
(7)		DBStd	12.09	6.01	6.55	5.35	4.86	3.74	3.45	3.28	3.63	4.32	4.97	5.40	5.78	(7)
(8)		HDD10.0	1300	386	278	91	12	0	0	0	1	29	152	351		(8)
(9)		HDD18.3	2776	644	510	304	111	11	1	0	1	21	179	386	609	(9)
(10)		CDD10.0	2676	0	2	46	174	369	503	572	541	334	116	17	0	(10)
(11)		CDD18.3	1111	0	0	1	23	123	254	313	283	105	8	1	0	(11)
(12)		CDH23.3	15014	0	1	17	258	1417	3510	4582	3873	1226	127	3	0	(12)
(13)		CDH26.7	8278	0	0	2	69	601	1993	2778	2263	550	22	0	0	(13)
(14)	Wind	WSAvg	2.3	1.8	2.1	2.5	3.0	3.0	2.6	2.5	2.7	2.3	1.9	1.6	1.6	(14)
(15)	Precipitation	PrecAvg	203	21	23	28	29	23	7	3	1	2	11	22	29	(15)
(16)		PrecMax	401	48	74	79	71	74	45	24	18	17	55	55	68	(16)
(17)		PrecMin	103	4	0	3	0	0	0	0	0	0	0	1	1	(17)
(18)		PrecStd	59	10	16	16	21	20	10	5	3	3	12	15	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	17.2	25.4	31.3	36.0	40.3	42.3	41.2	37.0	29.3	23.4	12.7	(19)
(20)			MCWB	7.4	10.1	13.6	16.8	18.0	19.8	19.8	19.7	17.9	14.8	13.2	7.5	(20)
(21)		2%	DB	9.3	13.5	22.3	28.6	33.8	38.3	40.1	39.0	34.4	26.5	18.9	9.8	(21)
(22)			MCWB	5.7	8.1	12.6	15.7	17.2	18.8	19.5	19.3	17.0	13.6	11.6	6.4	(22)
(23)		5%	DB	6.7	10.8	19.4	26.4	32.0	36.7	38.2	37.4	32.2	24.5	15.7	7.8	(23)
(24)			MCWB	4.1	6.8	11.4	14.8	16.5	18.5	19.1	18.8	16.2	12.9	9.6	5.1	(24)
(25)		10%	DB	4.7	8.3	16.8	24.1	30.1	35.1	36.6	35.9	29.9	22.0	13.4	5.9	(25)
(26)			MCWB	2.9	4.9	10.5	13.8	16.1	18.1	18.7	18.2	15.1	11.8	8.4	3.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	11.1	15.4	18.6	19.8	21.2	21.9	21.5	19.0	16.0	14.4	8.8	(27)
(28)			MCDWB	11.4	15.9	21.2	26.8	32.3	35.1	35.5	37.0	34.5	26.7	20.4	11.0	(28)
(29)		2%	WB	6.1	9.0	13.7	17.2	18.7	20.2	20.9	20.5	17.5	14.6	12.5	6.9	(29)
(30)			MCDWB	8.6	12.8	20.3	25.2	29.8	33.9	35.4	35.7	32.3	23.9	17.8	9.0	(30)
(31)		5%	WB	4.5	7.0	12.4	16.0	17.9	19.6	20.2	19.6	16.6	13.5	10.6	5.4	(31)
(32)			MCDWB	6.4	9.9	18.0	24.0	28.4	33.1	34.6	34.9	30.6	22.6	15.1	7.6	(32)
(33)		10%	WB	3.0	5.3	11.1	14.8	17.1	18.9	19.6	18.8	15.7	12.3	8.8	3.9	(33)
(34)			MCDWB	4.6	8.0	16.1	22.3	27.0	32.2	33.8	34.0	28.9	20.9	12.5	5.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	9.0	11.1	11.6	13.0	14.0	14.1	14.3	14.4	13.5	9.9	7.8	(35)
(36)			MCDBR	9.5	12.2	14.3	14.5	14.9	15.3	15.2	15.6	16.2	16.1	13.1	10.1	(36)
(37)			MCWBR	6.7	7.5	7.2	6.1	5.3	5.3	5.0	5.3	6.3	7.5	7.5	7.0	(37)
(38)		5% WB	MCDBR	8.3	10.6	12.2	12.9	13.2	14.2	14.1	15.0	15.3	14.5	11.4	8.7	(38)
(39)			MCWBR	6.3	7.2	6.8	6.0	5.2	5.1	5.0	5.5	6.4	7.2	6.8	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.360	0.393	0.451	0.481	0.466	0.451	0.436	0.403	0.380	0.407	0.402	0.357	(40)	
(41)		taud	2.058	2.033	1.967	1.938	2.022	2.093	2.145	2.222	2.239	2.078	2.052	2.085	(41)	
(42)		Ebn at Noon	723	770	774	787	813	826	833	851	842	745	666	684	(42)	
(43)		Edh at Noon	108	132	161	179	169	158	148	133	121	125	109	97	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.64	2.65	3.85	5.43	6.89	7.66	7.53	6.80	5.34	3.43	1.99	1.41	(44)	
(45)		RadStd	0.22	0.26	0.29	0.51	0.41	0.32	0.30	0.21	0.21	0.24	0.20	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.89	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	+183	+133

Nomenclature: See separate page